

Team Contributions

The project was completed collaboratively by all five team members. Responsibilities were divided across the major phases of the project, including literature review, data collection, mathematical modelling, computational implementation, testing, documentation, and project management. GitHub Issues, commits, pull requests, and team meetings were used to coordinate development.

Team Member	Primary Responsibilities
Emmanuel Ezenwata 1311369	Project coordination, GitHub repository management, project board and issues, mathematical model development, integration of project documents, README preparation, model justification, testing of computational results, final documentation review.
Bárbara Calderón Ávila 50281343	Literature review, academic source analysis, data collection and analysis, support for modelling decisions, report writing, proofreading, and reference management.
Jackson Mwema 50141199	Data collection, Google Maps distance and travel-time matrix preparation, customer demand scenarios, traffic modelling, and validation of input datasets.
Marko Antonio García Márquez 50281439	Python implementation, Google OR-Tools optimisation model, Clarke-Wright Savings Algorithm implementation, solver debugging, computational experiments, and performance analysis.
Abdessamad Amali 50152834	Data Collection, result validation, route analysis, visualisations, presentation preparation, testing of optimisation scenarios, and quality assurance of project outputs.

Detailed Contributions (Final Report Version)

Emmanuel Ezenwata

- Coordinated the overall project workflow.
- Managed the GitHub repository, Issues, milestones, and project board.
- Integrated the mathematical formulation, model justification, README, and project documentation.
- Reviewed consistency between the reports, implementation, and computational results.
- Supported validation of optimisation outputs.

Bárbara Calderón Ávila

- Conducted the literature review and data collection.
- Summarised and analysed academic papers related to VRP, CVRP, TDHVRPTW, heterogeneous fleets, and time windows.

- Supported the justification of modelling assumptions.
- Assisted with report writing, editing, and reference formatting.

Jackson Mwema

- Collected and organised operational data.
- Prepared customer demand datasets.
- Generated Google Maps distance and travel-time matrices.
- Assisted with traffic congestion assumptions and scenario preparation.
- Verified the consistency of input data.

Marko Antonio Garcia Marquez

- Implemented the optimisation model in Python.
- Developed the Google OR-Tools routing implementation.
- Implemented the Clarke-Wright Savings heuristic.
- Performed debugging, optimisation testing, and computational experiments.
- Generated routing outputs for all scenarios.

Abdessamad Amali

- Collected and organised operational data.
- Researched relevant literature related to TDHVRPTW and routing-model assumptions.
- Analysed optimisation results, route feasibility, and scenario outputs.
- Prepared figures, charts, route visualisations, and presentation material.
- Performed final quality assurance by checking consistency between README, Results, Docs, source-code filenames, and archive files.

Contribution Matrix

Project Activity	Emmanuel	Bárbara	Jackson	Marko	Abde
Project planning	✓	✓	✓	✓	✓
Literature Research					✓
Literature review	✓	✓			
Data collection		✓	✓		✓
Traffic modelling	✓		✓		
Mathematical formulation	✓	✓		✓	
Python implementation			✓	✓	
Clarke-Wright implementation				✓	
OR-Tools implementation				✓	
Computational testing	✓		✓	✓	✓

Results analysis	✓	✓		✓	✓
README and documentation	✓	✓			✓
Final repository quality assurance	✓				✓